

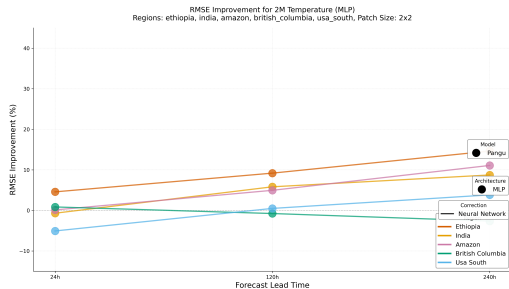
Scratch Work: Tailoring Machine Learning Weather Predictions for Local Impacts

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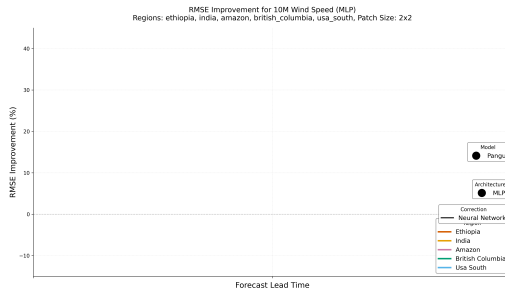
What's new?

- Rewrote databuild to be much faster and allow for using more regions. Early results for Ethiopia
- Added month embeddings to unet and mlp architectures.
- Capable to train on multiple lead times at once using learned lead time embedding.
- Implemented Bayesian hyperparameter optimization and optimized parameters for 2m temperatures 1 week lead time in India.

Do Better with Temperature: 2x2

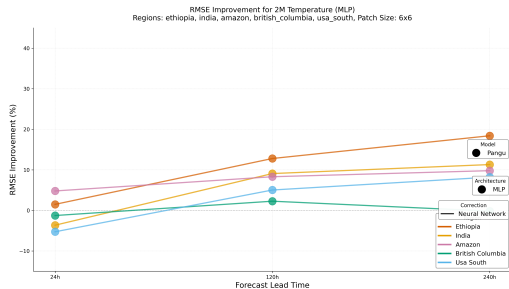


(a) 2m Temperature: Pangu

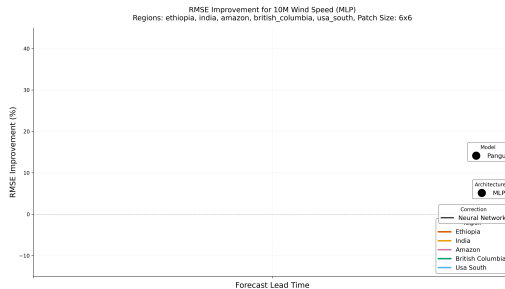


(b) Wind Speed: Pangu

6x6 Region Size

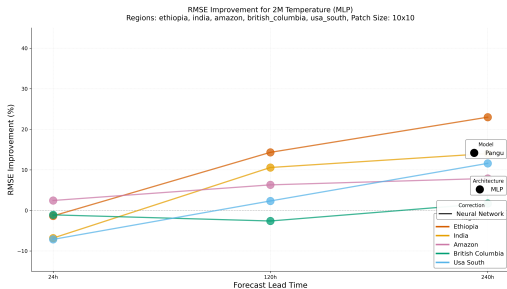


(a) 2m Temperature: Pangu

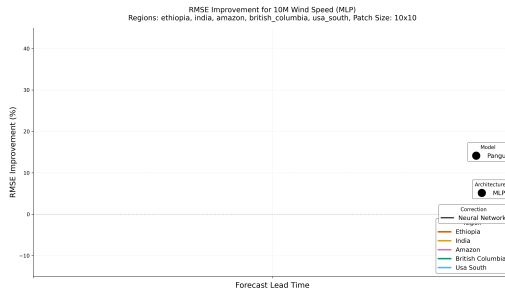


(b) Wind Speed: Pangu

10x10 Region Size



(a) 2m Temperature: Pangu



(b) Wind Speed: Pangu

Summary Table: 2m Temperature

Region	Lead Time	RMSE	Improvement (%)
Usa South	24h	1.320 (+0.094)	-7.1
<i>Ground Truth: 288.88 (12.31)</i>	120h	2.559 (-0.060)	2.3
	240h	5.454 (-0.633)	11.6
India	24h	0.917 (+0.063)	-6.8
<i>Ground Truth: 299.72 (6.79)</i>	120h	1.453 (-0.154)	10.6
	240h	2.013 (-0.282)	14.0
Amazon	24h	0.848 (-0.021)	2.4
<i>Ground Truth: 298.23 (1.62)</i>	120h	1.034 (-0.065)	6.3
	240h	1.291 (-0.102)	7.9
British Columbia	24h	1.602 (+0.018)	-1.1
<i>Ground Truth: 272.22 (13.42)</i>	120h	2.787 (+0.073)	-2.6
	240h	6.008 (-0.103)	1.7
Ethiopia	24h	1.080 (+0.015)	-1.4
<i>Ground Truth: 208.31 (7.27)</i>	120h	1.518 (-0.217)	14.3

Summary Table: 10m Wind Speed

Region	Lead Time	RMSE	Improvement (%)
Usa South	24h	1.012 (-0.068)	6.7
<i>Ground Truth: 3.97 (2.05)</i>	120h	1.529 (-0.164)	10.7
	240h	2.298 (-0.447)	19.5
India	24h	0.658 (-0.055)	8.3
<i>Ground Truth: 2.48 (1.41)</i>	120h	0.865 (-0.087)	10.1
	240h	1.088 (-0.131)	12.0
Amazon	24h	0.584 (-0.101)	17.2
<i>Ground Truth: 1.36 (0.65)</i>	120h	0.671 (-0.135)	20.2
	240h	0.779 (-0.164)	21.0
British Columbia	24h	0.558 (-0.067)	12.0
<i>Ground Truth: 2.10 (1.10)</i>	120h	0.859 (-0.106)	12.4
	240h	1.252 (-0.269)	21.5
Ethiopia	24h	0.703 (-0.073)	10.4
<i>Ground Truth: 2.61 (1.84)</i>	120h	0.808 (-0.164)	18.3